Tracking Error as an Observation for Behavior Cloning on Low-Cost Arms

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Abstract: Behavior-cloning stacks for low-cost arms expose one timestep of measured joint positions to the policy. We show that appending the servo tracking error, the difference between the last applied position command and the position reached, changes what a policy can learn at a 50-demonstration budget. On a physical SO-101, ACT policies with tracking error place the object in 23 of 40 paired trials against 6 of 40 for position-only policies (stratified exact test $p = 0.0005$), with the gain replicated across two training seeds. In simulation over five seeds, placement rises from 6.2% to 21.8%, and a position-history control recovers most of that gain, so the effect is attributable to temporal information the default omits rather than to tracking error specifically. The signal is free: it is reconstructible from any log that records commands alongside positions, and it keeps varying where the servo's load register saturates at a vendor torque limit in 16.5% of demonstration frames. Across 546 archived teleoperation episodes, tracking error at grasp onset marks resistance but not grasp success, so it is an observation, not an outcome label.

Keywords: Behavior Cloning, Proprioception, Low-Cost Manipulation

1 Introduction

The proprioceptive input to a behavior-cloning policy is rarely a decision. ACT [1], the LeRobot stack [2], and the vision-language-action models built on them read the current measured joint positions, one timestep, nothing else. On a hobby-servo arm the omission removes the contact channel: the servos are proportional position controllers, so when a jaw meets an object the measured position stalls while the command deepens, and the gap between them is the only record of the load. The stack keeps one operand of that subtraction in the action labels and never provides the difference as an observation.

This paper treats that default as an object of study, in the way recent work has treated controller gains and action spaces [3, 4]. We append the tracking error, $\delta_t = g_{t-1} - q_t$, to the observation of an otherwise unchanged ACT policy and measure what changes. The feature costs nothing: it needs no sensor, no force model, and no calibration, and it can be reconstructed after the fact from any recording in which the logged commands are the goals the servos received. Tracking error also mixes contact with following lag and actuator dynamics, and a short history of positions carries much of the same information. The questions are whether exposing it helps, whether the help is specific to the command channel, and what a paper should report so that the answer transfers.

We answer with three experiments. A simulation grid sets explicit tracking error against action history, position history, and a compensated residual under identical data and training budgets, five seeds each. A hardware study trains ACT on 50 human demonstrations on one SO-101 and evaluates 80 paired trials across two training seeds. A corpus study asks, on 546 archived episodes, whether the same signal can serve as an outcome label. Tracking error wins both hardware comparisons, by 4 and 13 placements out of 20. In simulation, position history matches it. The result is a statement about the interface, not about force estimation: one timestep of position is the wrong default on this hardware, any temporal channel repairs it, and tracking error is the one every existing log already contains.

2 Related work

Hwang et al. [5] estimate torque on RC servos by system identification, FACTR 2 [6] learns external-torque estimates from free-motion data and resamples training data by force, and NeuralActuator [7] fits an actuator model on the SO-101 from command, position, tracking error, and the load register and feeds its estimated force to a policy. We use the raw command–position difference with no model and make correspondingly weaker claims about force. Force-aware imitation on sensed arms finds that how the signal is presented matters [8, 9].

Wong et al. [10] show that leader–follower discrepancies in teleoperation carry interaction cues: removing them from ACT's action labels degrades force-critical phases, while torque proxies restore contact-aware behavior, and Xu et al. [11] condition policies on the mismatch history. Our intervention adds the discrepancy to the observation of an unchanged policy, which makes this an observation-design study rather than a sensorless-force principle. Adding the feature also adds access to the previous command, and history that helps under partial observability can equally support a shortcut that copies the previous action [12, 13, 14]. We therefore compare several temporal observations rather than reading any gain as contact sensing.

3 Observation and experimental design

Let q_t be the measured joint position and g_{t-1} the preceding applied position goal, expressed in the same coordinate system. We define

$$\delta_t = g_{t-1} - q_t, \qquad o_t^{\text{base}} = [I_t, \widetilde{q}_t], \qquad o_t^{\delta} = [I_t, \widetilde{q}_t, \delta_t/c], \tag{1}$$

where I_t is the camera observation and $\widetilde{q}_t$ uses training-set normalization. The scale $c = 8$ is fixed in each dataset's joint-coordinate units and is not a conversion to torque. Under approximately static contact, tracking error reflects load on a position servo; during motion it also contains following error, and we use it as an observation without assigning every nonzero value to contact.

Simulation. We train ACT policies [1] on 280 scripted pick-and-place demonstrations with randomized object and scene properties. All designs share the same data and training budget: 12,000 steps with 96-pixel front and gripper-camera images. Each policy is evaluated on 100 episodes with shared scene seeds across designs, with five training seeds for the original designs and three for the position-history control. The comparison uses placement success with the crush threshold disabled, so it tests grasping and transport rather than regulation within a force band.

Hardware. We collect 50 teleoperated demonstrations of placing an adapter inside a tape roll using one SO-101 and one overhead camera. Both policies use the same 18,549 training frames after removal of stationary prefixes, 96-pixel images, and 6,000 training steps. The proprioceptive input has six dimensions for the base and twelve with tracking error; every other architecture and optimization setting is shared. Three seeds are trained, one checkpoint per observation design and seed, and every checkpoint is evaluated.

Evaluation begins after replaying 115 commands from a fixed demonstration, which places the arm near the grasp, after which each policy controls up to 400 steps. In the demonstrations the grasp closure occurs at a median frame 331 of a median 400-frame episode and after frame 115 in 48 of 50 episodes, and a median 67% of joint travel follows frame 115, so the replay covers the approach and leaves the grasp to the policy (Appendix B). ACT predicts and executes 30-command chunks, so observations inform the next chunk rather than each command within it. A relative-target limiter is shared by both policies, and the tracking-error input uses the command returned *after* limiting. Trials are paired by training seed and physical reset, with base-first and tracking-error-first order alternating across pairs, and an operator records whether the adapter is placed inside the roll (Appendix A).

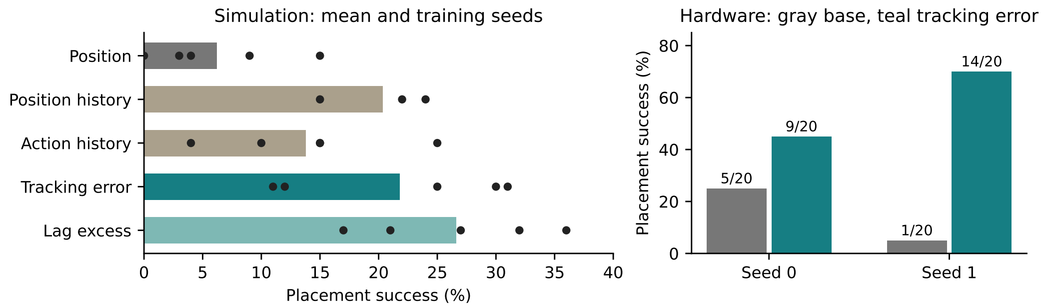

Figure 1: Placement success by observation design in simulation and in the two completed hardware evaluations. Left: bars are means; dots are individual training seeds (five per design, three for position history). Right: hardware success counts for the two completed evaluations. The full evaluation record, including an interrupted third comparison, appears in Appendix A.

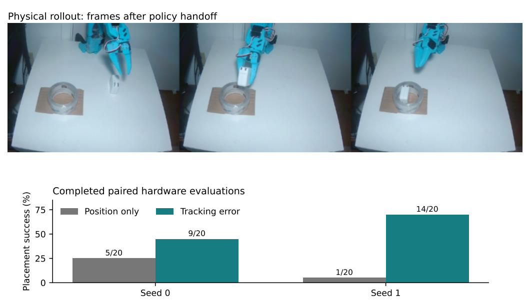

Figure 2: Top: three frames from a tracking-error policy (seed 1) after the scripted replay, from a showcase rollout that is not included in any count. Bottom: placement successes over 20 pairs for the two completed hardware evaluations.

4 Results

Hardware: a 4$\times$ improvement at 50 demonstrations. Table 1 reports each training seed as its own paired comparison. Tracking error improves placement by 20 percentage points for seed 0 and by 65 points for seed 1. Exact two-sided McNemar tests use only the discordant pairs: seed 0 has six pairs favouring tracking error and two favouring the base ($p = 0.29$), seed 1 fourteen and one ($p = 0.001$). Pooling the two seeds gives 23 of 40 placements against 6 of 40, with 20 discordant pairs favouring tracking error against 3 (stratified exact test $p = 0.0005$; conditional odds ratio 6.7, 95% CI 2.0 to 35). These tests describe trial-to-trial variation for fixed checkpoints, not variation over training seeds; with 20 pairs the exact test has power 0.17 under seed 0's observed discordance structure and 0.97 under seed 1's, so each comparison alone can detect only large effects (Appendix A). A third evaluation was interrupted by recurring gripper faults after six all-failure pairs for both policies; its partial results and missing outcomes are reported in Appendix A.

The checkpoints read the channel: zeroing the tracking-error input offline moves the predicted jaw command by 12.6 to 13.0 units on frames where the gripper load register is pinned and by 2.7 to 2.9 units elsewhere, against a median recorded command step of 0.5 (Appendix C). The dependence concentrates where the jaw is loaded, which is where the signal is informative, without showing that it produced the placements.

Simulation: the default is the problem, not the feature. Mean placement success is 6.2% for the position-only base, 13.8% with the previous action appended, 21.8% with explicit tracking error, and 26.6% with a compensated residual (Fig. 1). Tracking error exceeds the base by 15.6 points (Welch 95% CI 3.5 to 27.7), and both it and the residual clear the preregistered resolution rule of $15\sqrt{3/n}$ points for n seeds. A position-history control that observes $[q_t, q_{t-8}]$ and no commands at

Table 1: Hardware placements and paired outcomes for the two completed evaluations. Discordances list tracking-error-only / base-only successes. See Appendix A for the interrupted evaluation.

Seed	Evaluation	Base	Tracking error	Discordances	p
0	Complete	5/20	9/20	6 / 2	0.2891
1	Complete	1/20	14/20	14 / 1	0.0010

all reaches 20.3% over three seeds, 14.1 points above the base (CI 4.5 to 23.8) and 1.5 points below tracking error (CI -11.1 to 14.0). Temporal information is missing from the default, restoring it roughly triples placement, and its form matters less than its presence, since a pure position history recovers most of the gain. Whether explicit tracking error is preferable to a suitable history is not resolved at this seed count (Appendix F). Predicting tracking error as an auxiliary training target gives 6.6%: the channel helps as an input, not as a target.

The vendor torque limit clips the shipped force channel. The SO-101's servos expose a load register that the stack does not read. In the 50 demonstrations that register pins at its configured limit of 500 in 16.5% of frames, while tracking error keeps varying across those same frames (Appendix D); over all frames the two channels correlate at $r = -0.92$, and on a static bench with hanging masses the register is a linear function of tracking error ($R^2 = 0.976$). The reduced torque limit in the inspected follower configuration is the default that produces the plateau. Tracking error is thus not only the channel every log already has; on this hardware it is the one that survives the vendor default.

Corpus: an observation, not an outcome label. A preregistered prediction held that larger tracking error at detected grasp onset would accompany successful grasps in archived teleoperation data. Sixteen public datasets with 546 episodes pass the inclusion gates. Only two meet the predicted effect-size threshold ($d > 0.5$); the pooled within-dataset effect is negative ($d = -0.39$) against a gripper-retention proxy. Empty closure against a mechanical stop and an operator closing harder against resistance both produce large discrepancy, so the signal marks resistance without identifying a successful grasp (Appendix G). This is the expected behavior of a contact channel and the reason it belongs in the observation rather than in a labeling rule.

5 Discussion and conclusion

On a physical low-cost arm, an ACT policy given the servo tracking error places the object four times as often as the same policy given positions alone, at a 50-demonstration budget and across two training seeds. In simulation the same gain appears and a position-only history recovers most of it, so the operative defect is the single-timestep proprioceptive default, not a missing sensor. The channel that repairs it is already in every recording that logs commands, and it stays informative after the vendor torque limit has flattened the servo's own load register.

This suggests three reporting items for the substrate beneath the policy. State the proprioceptive observation, including its history depth, since the default is invisible and decisive. Log applied commands, after any limiter, alongside measured positions, since that pair is the contact channel on position-controlled hardware. Report actuator torque and current limits, since they set where the shipped telemetry saturates.

Three limits bound the claim. The two hardware seeds gave a 20-point and a 65-point improvement and two seeds cannot bound that spread. The matched position-history control is registered, and its first hardware session was interrupted by a gripper servo fault (Appendix H), so the hardware result establishes the value of the channel, not its superiority over history. The hardware task never requires staying under a force threshold, so nothing here shows learned force regulation. More seeds, the hardware history control, and a force-limited task are the next experiments.

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A Hardware protocol and complete evaluation record

Table 2: Full hardware evaluation record: success counts with Wilson 95% intervals in percentage points, discordant and concordant pair counts, exact McNemar p per seed, and the stratified exact test over the completed seeds. The interrupted seed has six recorded pairs, with 0/6 placements for each policy; its planned budget was 20 pairs. No confirmatory test is assigned to the incomplete comparison.

Seed	Evaluation	Base [95% CI]	Tracking error [95% CI]	TE / base only	both / neither	p
0	Complete	5/20 [+11, +47]	9/20 [+26, +66]	6 / 2	3 / 9	0.2891
1	Complete	1/20 [+1, +24]	14/20 [+48, +85]	14 / 1	0 / 5	0.0010
2	Interrupted	0/6 [+0, +39]	0/6 [+0, +39]	0 / 0	0 / 6	—
0+1	Stratified			20 / 3		0.0005

Table 3: Power of the exact two-sided McNemar test with 20 pairs at $\alpha = 0.05$, by the probability that a pair is discordant and the probability q that a discordant pair favours tracking error. Seed 0's observed structure (8 discordant pairs, 6 favouring) gives 0.17; seed 1's (15, 14) gives 0.97.

P(discordant)	$q = 0.7$	$q = 0.8$	$q = 0.9$	$q = 1.0$
0.2	0.02	0.05	0.10	0.20
0.4	0.11	0.26	0.53	0.87
0.6	0.19	0.46	0.80	1.00
0.8	0.29	0.64	0.93	1.00

Training. The recording contains 21,476 frames from 50 demonstrations at 30 Hz. Position and action statistics are computed on the recording; the training index then removes stationary prefixes, retaining 18,549 frames. Both policies use the same index and normalization, and recorded load channels are excluded from policy inputs. Gripper position is in the interface's own units, not a calibrated force.

The ACT configuration uses a ResNet-18 image backbone, model width 256, four attention heads, feedforward width 1024, two encoder layers and one decoder layer. The action chunk and execution horizon are both 30 steps, with no temporal ensembling. Optimization uses AdamW at learning rate 10^{-4}, weight decay 10^{-4}, batch size 8 and gradient clipping at 10. All reported checkpoints are the final checkpoint at 6,000 steps. The observation projection necessarily changes width when six features are added; the remaining architecture and training settings are shared.

Execution and command semantics. Each rollout homes the arm to the mean demonstration start, replays the first 115 commands from demonstration 0, and hands control to the policy, so the reported task excludes autonomous approach from an arbitrary initial state. Each requested command passes through a relative-target limit of 8 in the interface's joint units—a target-displacement bound rather than a force or velocity guarantee—and during both replay and policy execution the returned limited command supplies the next tracking-error calculation. The servo's native goal quantization sits below this software interface. State and images are read at each control step, but ACT consults them only when its 30-command queue is empty.

An offline replay check reproduces the training and inference state vectors on all 21,476 recorded frames for the base, tracking error, action history, position history and compensated residual designs. In seed 0, the saved tracking-error observations after the first rollout step also reproduce from logged applied commands, which checks software consistency, not physical command-to-encoder latency. Collection-time command limiting has not been established from the available metadata, so archived action labels should be read as recorded goals, which need not equal the goals applied to a servo if collection used an additional limiter.

Pairing and inference. Hardware resets are manual, and within each seed consecutive trials form a pair with AB/BA order alternating between pairs. The operator knows the policy identity, so outcomes are not blinded. Outcomes show no order effect: seed-1 tracking error placed 7/10 whether

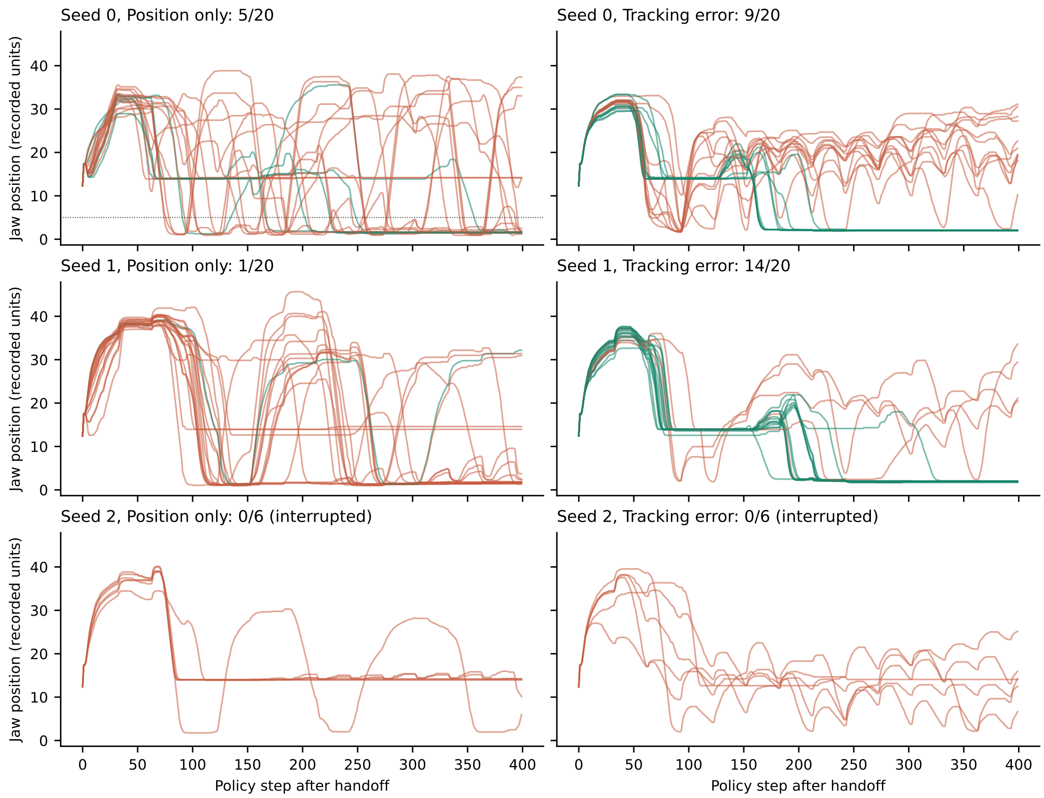

Figure 3: Recorded jaw position for every hardware rollout, by seed and policy; green traces placed the adapter, orange did not. Policy control begins at step 0 after the scripted replay. The dotted line marks the closure threshold used in Table 4.

it ran first or second in a pair and in both halves of the session, and seed 0 placed 5/10 and 4/10 by order and 3/10 and 6/10 by half. Seeds 1 and 2 were scheduled after observing seed 0, with their training recipe and evaluation budget fixed before their evaluations, and no policy was selected by hardware performance: every trained checkpoint was evaluated. The primary paired test is exact, two-sided McNemar with an improvement criterion of positive effect and $p < 0.05$; for the completed seeds, paired percentile bootstrap intervals (20,000 pair resamples, random seed 0) are $[-5, +45]$ and $[+40, +90]$ points, describing trial uncertainty for fixed checkpoints, not confidence intervals over retraining.

Interruptions and missing data. Seed 0 has 40 outcomes and 40 trajectories. Seed 1 has 40 outcomes and 39 trajectories: two rollouts, one per policy, completed their control horizon but reported a gripper overload while torque was being disabled; both are operator-labeled failures, and one trajectory file was lost to a logging bug that was fixed before evaluation resumed. Seed 2 was stopped after twelve labeled rollouts, all failures, forming six pairs; three reported the same overload at shutdown, and a thirteenth rollout completed its horizon but its outcome was not entered. We leave that outcome missing rather than infer it from joint motion, and we have no diagnosis of the recurring overload. In the seed-2 rollouts the jaw never closed in five of six position-only and three of six tracking-error trials, against three of 20 and one of 20 in seed 0 and two of 20 and none of 19 in seed 1 (Table 4), which is consistent with a gripper fault rather than a policy difference, although the traces cannot prove it. Trial-level detail is in the repository's evaluation records.

Table 4: Jaw-trace classes for all recorded rollouts. Never closed: jaw never below 5 recorded units. Reopened: an excursion above 8 units after closing. Closure step is the median first step below the threshold. One seed-1 tracking-error trajectory is missing.

Seed	Policy	Outcome	n	never closed	closed, reopened	closed, held	median step of closure
0	Position only	success	5	0	1	4	203
0	Position only	failure	15	3	8	4	110
0	Tracking error	success	9	0	0	9	165
0	Tracking error	failure	11	1	10	0	83
1	Position only	success	1	0	1	0	127
1	Position only	failure	19	2	13	4	116
1	Tracking error	success	14	0	0	14	206
1	Tracking error	failure	5	0	4	1	115
2	Position only	failure	6	5	1	0	88
2	Tracking error	failure	6	3	3	0	241

B Exploratory trajectory analysis

Every recorded rollout was classified from its jaw trace as never closed, closed then reopened, or closed and held (Table 4, Fig. 3). All 23 tracking-error successes in seeds 0 and 1 closed once and held to the end of the horizon, and 15 of their 16 failures either never closed or reopened. Position-only successes are too few to characterize. Successful tracking-error rollouts closed later, at median step 165 and 206 against 83 and 115 for failures, and in seed 0 they requested smaller moves: the median fraction of steps altered by the command limiter is 0.09 for tracking error and 0.32 for the base, while in seed 1 the fractions are 0.32 and 0.35. Over the last 100 steps of the seed-0 tracking-error rollouts, median summed joint-coordinate travel, a descriptive sum over mixed coordinates, is 14.3 for successes and 93.6 for failures. Later closure and less motion in successful trials could simply follow from progressing further through the task. Traces were grouped by outcome after the fact and contain no video, object pose or force, so they cannot say whether a failure was a missed grasp, a slip or a misplacement, and none of these regularities is a validated success detector. In tracking-error rollouts the median absolute jaw tracking error after closure is 0.11 and 0.16 units for successes and 1.04 and 0.88 for failures (seeds 0 and 1): a held grasp settles to a small steady discrepancy, whereas reopening and re-closing keeps it large. Fig. 4 shows when the jaw first closes in each rollout and, for the demonstrations, when the grasp closure occurs relative to the replayed prefix.

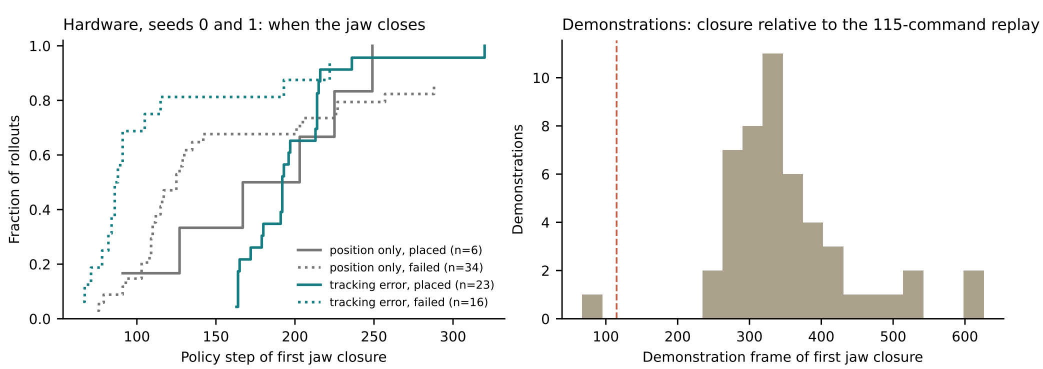

Figure 4: Left: cumulative fraction of hardware rollouts (seeds 0 and 1) whose jaw has closed by a given policy step, by policy and outcome. Right: frame of the grasp closure in the 50 demonstrations; the dashed line is the 115-command replay.

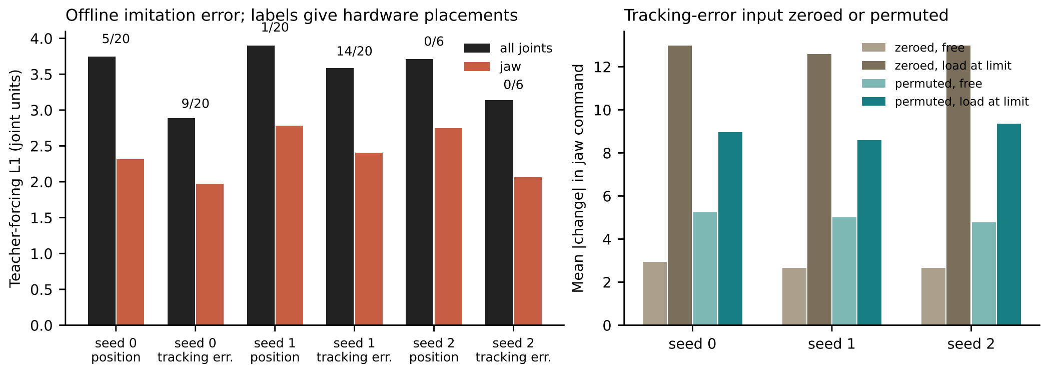

Figure 5: Left: teacher-forcing L1 error of each evaluated checkpoint's predicted 30-command chunk on every fourth training frame (4,638 frames), all joints and jaw; labels give hardware placements. Right: mean absolute change in the predicted jaw command when the six tracking-error inputs are zeroed or permuted across frames, split by whether the recorded gripper load register is at its limit (881 frames) or not.

C Checkpoint sensitivity and offline error

The hardware trials cannot say whether a policy reads its tracking-error input, so the six evaluated checkpoints were re-scored offline on the demonstration frames (Fig. 5). Zeroing the input changes the predicted jaw command by 12.6 to 13.0 units where the load register is pinned and by 2.7 to 2.9 units on free frames; permuting it across frames changes the command by 8.6 to 9.4 and 4.8 to 5.2 units. The median recorded command step is 0.5 units, and teacher-forcing error rises from 2.9 to 3.6 to 6.2 to 6.8 under zeroing. All three tracking-error checkpoints therefore depend on the channel, most where the jaw is loaded; this is a property of the networks, not evidence that the dependence is what produced the hardware placements. Offline error itself is a fit diagnostic on training frames, since no demonstrations were held out: it separates designs but ranks the seed-0 tracking-error checkpoint (2.9, 9/20) below the seed-1 checkpoint (3.6, 14/20). Training losses of the seed-1 and seed-2 checkpoints track each other closely (Fig. 6).

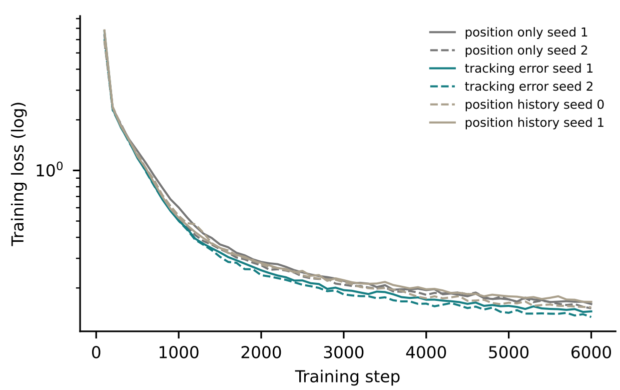

Figure 6: Training loss, logged every 100 steps, for the checkpoints whose logs are in the repository.

D The shipped load register saturates at a vendor limit

A separate static experiment on one shoulder-lift STS3215 uses twelve hanging masses at five poses (60 measurements). Across those measurements, the fitted relationship between the load register and tracking error is `Present_Load` $= -4.70\delta + 0.1$, with $R^2 = 0.976$. Its relation to applied

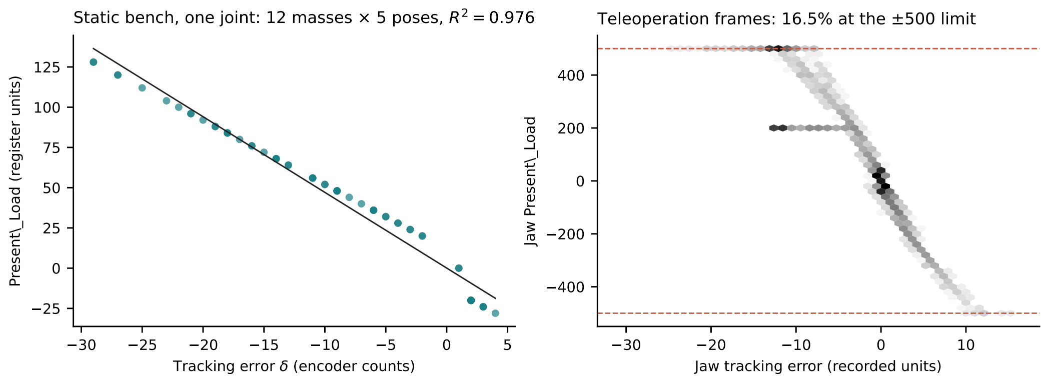

Figure 7: Left: static bench, one shoulder-lift servo, load register against tracking error over twelve hanging masses at five poses, with the fitted line. Right: jaw load register against jaw tracking error on the teleoperation frames (log-count shading); the dashed lines are the configured ± 500 torque limit.

mass is far less stable, with per-pose R^2 for tracking error ranging from 0.085 to 0.910, so the tight load-register association does not establish an equally accurate physical-load measurement.

In the 50 teleoperation demonstrations, the gripper load register reaches 500 in every episode and remains there in 16.5% of frames, while tracking error continues to vary across those same frames (Fig. 7); over all frames the two channels correlate at $r = -0.92$. The inspected follower configuration sets a reduced gripper torque limit, consistent with that observed plateau. This motivates inspecting the command–position discrepancy when telemetry saturates, but it does not calibrate the remaining discrepancy in newtons: no jaw load-cell calibration is available, and simulator force conversions such as the fitted 2.00 N/count slope and 4.3 N held-out RMSE over 4–78 N describe the simulated plant only.

E Simulated plant

The simulated plant was characterised with scripted experts before any policy was trained, to check that tracking error can carry contact information in this simulator at all. Three experts differ only in the signal that sets the grip: a fixed jaw angle (clamp), the quantised tracking error, or the true contact force (oracle). Without a crush limit the clamp places 29 of 30 and the two regulated experts 22 and 20, because a gentler grip drops more often and nothing penalises crushing. With a crush limit the clamp fails every episode at a median peak grip of 355 N, while the tracking-error expert at the native quantum places within a few episodes of the oracle at every limit, and coarsening the observation quantum destroys the task as a cliff whose position moves with the margin (Table 5). This is why the learning grid could ask whether a policy uses the channel: in this plant the channel suffices for force regulation by a scripted controller at the real encoder's resolution. It does not show that the physical servo provides the same information; Appendix D addresses that separately.

Table 5: Scripted-expert placements out of 30 randomised episodes in the simulated plant, by grip signal and observation quantum, with and without a crush limit that fails the episode if grip force exceeds it.

	tracking error, quantum q (counts)					clamp	oracle
Crush limit	1	4	8	16	32		
no limit	22	—	—	—	—	29	20
100 N	11	11	9	1	0	0	12
120 N	17	17	16	10	0	0	14
160 N	20	20	19	17	11	0	17

F Additional simulation and hardware comparisons

Simulation recordings use integer encoder counts at 15 Hz, obtained by subsampling the 30 Hz controller. The scripted teacher regulates the jaw using tracking error, and the successful demonstration set includes randomized grip depth and recovery perturbations. These collection choices are shared across observation designs, of which the position-history control observes $[q_t, q_{t-8}]$ and one-command history observes $[q_t, g_{t-1}]$. Neither comparison makes the simulation and hardware tasks identical: camera views, demonstrations and recording cadence all differ between the two settings.

The original simulation study includes an auxiliary tracking-error target and a latched seat token in addition to the designs in Fig. 1, with mean placement rates of 6.6% and 21.8% across five seeds; the seat token matches tracking error exactly. The compensated residual is $e_t = \delta_t - \widehat{K}(g_{t-1} - g_{t-2})$, where each joint's coefficient is fitted once on presumed free-motion demonstration frames. A two-command history control reaches 14.7% over three seeds; comparing it with the residual holds available raw information fixed but does not match input width. Under the original study's preregistered practical resolution rule, $15\sqrt{3/n}$ percentage points for n training seeds, neither this comparison nor the residual–position-history comparison resolves at three seeds, and the stronger feature-specific interpretation remains open. Table 6 lists the main design contrasts with Welch intervals; the Holm column adjusts across all 28 pairwise comparisons, and no contrast survives that adjustment. The preregistered contrasts against the base are the primary tests; the adjustment across every pair is reported for completeness.

Table 6: Simulation design contrasts in placement-success points, with Welch 95% intervals, unadjusted and Holm-adjusted p values across all 28 pairs, the preregistered resolution rule $15\sqrt{3/n}$ for the smaller seed count, and whether the difference clears it.

Contrast	Points	Welch 95% CI	p	Holm p	Rule	Clears
Position history − Position	+14.1	[+4.5, +23.8]	0.013	0.33	15.0	no
Action history − Position	+7.6	[-2.5, +17.7]	0.121	1.00	11.6	no
Tracking error − Position	+15.6	[+3.5, +27.7]	0.019	0.44	11.6	yes
Compensated residual − Position	+20.4	[+10.2, +30.6]	0.002	0.05	11.6	yes
Auxiliary target − Position	+0.4	[-7.1, +7.9]	0.903	1.00	11.6	no
Tracking error − Position history	+1.5	[-11.1, +14.0]	0.784	1.00	15.0	no
Compensated residual − Position history	+6.3	[-4.6, +17.1]	0.206	1.00	15.0	no
Tracking error − Action history	+8.0	[-4.9, +20.9]	0.188	1.00	11.6	no
Compensated residual − Two-command history	+11.9	[+2.2, +21.7]	0.025	0.49	15.0	no
Compensated residual − Tracking error	+4.8	[-8.1, +17.7]	0.413	1.00	11.6	no

An earlier hardware comparison, run before the tracking-error study, evaluated the compensated residual: the base placed 7/10 objects and the residual 0/10 (Fisher exact $p = 0.003$; Wilson intervals 40 to 89% and 0 to 28%; alternating unpaired trials with the same 115-command replay). The coefficients fitted from these real demonstrations differ markedly from the simulation fits, and a free-motion labeling heuristic includes contact frames. Both are plausible sources of mismatch, not a diagnosis. Coefficients above one do not by themselves invalidate the regression, and that runner also fed back requested rather than post-limit commands, so preprocessing and execution both differ from the tracking-error study.

The simulation task admits a crush threshold, but the learning grid does not activate it. A post-hoc screen of existing checkpoints at 120 and 60 N produces heavy crushing and near-floor success, failing the predicted advantage of the compensated residual. A separate runtime guard reduces crush events from 85 to 3 across 180 crush-tier episodes, while eliminating all 12 unguarded placements across its broader 270-episode comparison. Neither result is learned force regulation, and the guard was not used on hardware.

Larger-budget experiments do not establish a scaling advantage. An earlier 600-demonstration comparison of tracking error and one-command history reaches 57% and 60%, respectively, over two seeds at 224 pixels and 50,000 steps. A later 600-demonstration screen saved at 26,000 steps contains three base and three residual evaluations, each with 0/100 placements. These floor results are reported separately from the 280-demonstration grid; they support neither a positive scaling claim nor equivalence between observations.

G Archived teleoperation study

The preregistered corpus prediction was that tracking error at a detected seat event would be larger in proxy-successful episodes, with Cohen's $d > 0.5$ in at least ten of twenty targeted datasets. Sixteen datasets and 546 episodes pass the frozen gates, including 283 proxy successes under a success proxy of continued gripper closure through transport. Only two datasets exceed the positive threshold, and the pooled within-dataset effect is $d = -0.39$—a failure of the proposed direction and labeling rule, though not proof that no function of command and position history can predict outcome. Eleven of the sixteen point estimates are negative and seven of those intervals exclude zero; two are positive with intervals excluding zero (Fig. 8).

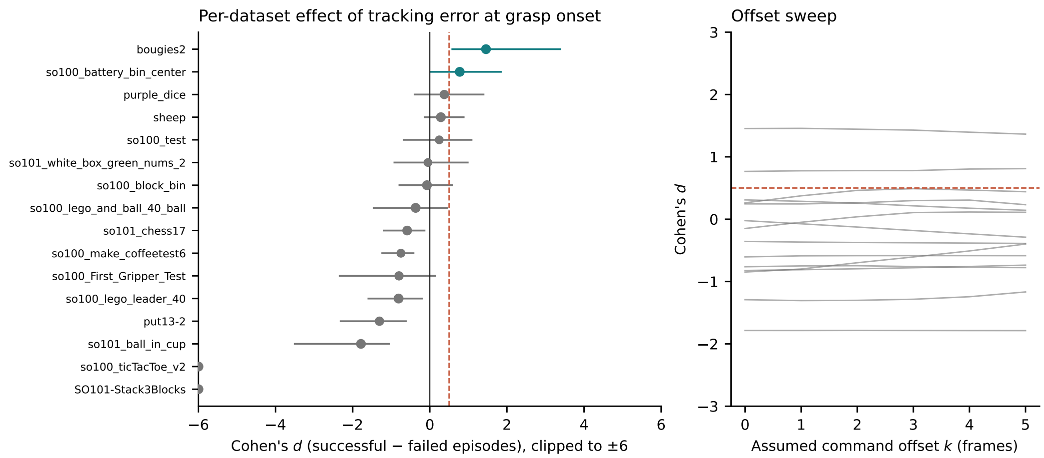

Figure 8: Left: per-dataset Cohen's d of tracking error at grasp onset between proxy-successful and failed episodes, with 95% intervals, clipped to ± 6; marker area scales with episode count and the dashed line is the preregistered $d = 0.5$ threshold. Right: the same effect sizes under assumed command offsets of zero to five frames.

Empty closure against a mechanical stop and closure on an object can both produce a seat event, and a large command discrepancy can equally reflect an operator continuing to close against resistance. These ambiguities explain why a contact-related feature need not be a success label, although the corpus does not isolate their causal contributions. Recorded leader commands may differ from applied follower goals and the timing is only approximately aligned, but an offset sweep from zero to five recorded frames leaves the qualitative conclusion unchanged, with median per-dataset effect-size shift 0.12.

H Position-history control on hardware

The simulation grid leaves open whether the hardware gain from tracking error is specific to the command or would follow from any temporal input. A matched position-history control, $[q_t, q_{t-8}]$ with the seed-1 and seed-2 training recipe, is pre-registered in the repository together with the two paired comparisons that would settle it: tracking error against position history for seed 1, and position only against position history for seed 0, each with 20 pairs under the protocol of Appendix A. The seed-1 comparison was run on 2026-09-07 and interrupted after 9 complete pairs (19 rollouts, 5 ending with a gripper overload error): the gripper servo stopped following jaw commands in several rollouts and the overhead camera was found displaced relative to the earlier sessions, so its outcomes are retained (Table 7) but the comparison is not complete and no confirmatory test is assigned to it. The other registered comparison has not been run.

Table 7: Pre-registered position-history comparisons on hardware. Generated from the trial records; empty until the trials exist.

Seed	Comparison	Session	First	Position history	history-only / first-only	p	faults
1	Tracking error vs. position history	Interrupted	1/9	0/9	0 / 1	1.0000	5